the centripetal force must act on the object radially towards the centre.

Let us consider an object of mass m, moving along a circular path of radius r, with a velocity, v. Its centripetal acceleration is given by

$$a = v^2/r$$

The magnitude of centripetal force is given by,

F = Mass × Centripetal acceleration

$F = mv^2/r$

Problem 4

A 900 kg car moving at 10 ms^{-1} takes a turn around a circle with a radius of 25 m. Determine the acceleration and the net force acting upon the car.

Solution:

When the car turns around circle, it experiences centripetal acceleration, $a = \frac{v^2}{r}$

$$a = \frac{(10)^2}{25} = \frac{100}{25} = 4\ ms^{-2}$$

Net force acting upon the car,

F = m a = 900 × 4 = 3600 N

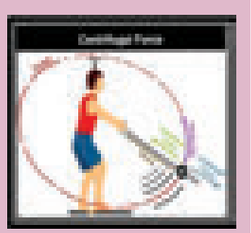
DO YOU KNOW?

Any force like gravitational force, frictional force, magnetic force, electrostatic force etc., may act as a centripetal force.

2.10 Centrifugal Force

Activity 6

Take a piece of rope and tie a small stone at one end. Hold the other end of the rope and rotate it such that the stone follows a circular path. Do you experience any pull or push in your hand?



In this activity, a pulling force that acts away from the centre is experienced. This is called as centrifugal force. Force acting on a body away from the centre of circular path is called centrifugal force. Thus, centrifugal force acts in a direction which is opposite to the direction of centripetal force. Its magnitude is same as that of centripetal force. The dryer in a washing machine is an example for the application of centrifugal force.

DO YOU KNOW?

When you go for a ride in a merry-go-round in amusement parks, you will experience an outward pull as merry-go round rotates about vertical axis. This is due to centrifugal force.

Points to Remember

- Motion is a change of position, which can be described in terms of the distance moved or the displacement.
- Motion of an object could be uniform or non-uniform depending on its velocity.
- Speed of an object is the distance covered per unit time and velocity is the displacement per unit time.
- The acceleration of an object is the change in velocity per unit time.
- The motion of an object at uniform acceleration can be described with the help of three equations, namely:

  $v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$
- For a freely falling body, the acceleration, a is replaced by g.
- An object under uniform circular motion experiences centripetal force.
- Centrifugal force acts in a direction which is opposite to the direction of the centripetal force.

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